

LEI CHANG

✉ changlei0406@163.com · ☎ (+86) 178-0029-0863

🎓 EDUCATION

University of Chinese Academy of Sciences

Sep. 2024 – Jun. 2027

Yinchuan Energy College

Sep. 2014 – Jun. 2018

👤 WORK EXPERIENCE

Bosch Automotive Products (Suzhou) Co., Ltd.

Apr. 2026 – Present

Voyah Automotive Technology Co., Ltd.

Nov. 2024 – Sep. 2025

NLP Algorithm Engineer

- Responsible for the research, development, and optimization of Large Language Models (LLMs), including supervised fine-tuning (SFT), reinforcement learning from human feedback (RLHF), and alignment algorithms such as PPO and DPO.
- Designed and optimized AI algorithms for intelligent cockpit systems, covering voice interaction, intent understanding, semantic parsing, and task execution.
- Participated in technical documentation and patent writing related to deep learning, reinforcement learning, and LLM applications in automotive scenarios.
- Built standardized model evaluation pipelines, including data collection, filtering, annotation, model iteration, and performance evaluation.
- Followed the latest advances in reinforcement learning and generative AI, and applied state-of-the-art techniques to real-world automotive applications.

Ultrapower Software Co Ltd.

Jul. 2021 – Apr. 2024

NLP Algorithm Engineer

- Developed and optimized Large Language Model algorithms and architectures by applying advanced techniques in NLP, deep learning, and reinforcement learning to improve generation quality and instruction-following capabilities.
- Designed intelligent cockpit AI solutions based on real-world automotive scenarios, including user intent recognition, semantic understanding, and dialogue management.
- Developed data processing pipelines for LLM training, including data collection, cleaning, filtering, annotation, augmentation, and instruction dataset construction.
- Built evaluation frameworks for generative AI models, analyzing model performance through automated metrics and human evaluation.
- Designed and deployed LLM-based applications using RAG, prompt engineering, and parameter-efficient fine-tuning methods.
- Conducted research on emerging AIGC technologies and transformed academic advances into practical engineering solutions.

China Automobile Industrial Engineering Co., Ltd.

Jul. 2018 – Jun. 2021

NLP Algorithm Engineer

- Designed AI algorithm architectures and technical solutions according to industrial requirements.
- Developed and optimized NLP algorithms for entity recognition, relation extraction, intelligent question answering, and semantic understanding.

- Built domain-specific NLP components, including tokenization, text summarization, classification, and information extraction modules.
- Conducted research on machine learning and deep learning methods and applied them to practical industrial AI systems.

⚙️ PROFESSIONAL SKILLS

- Proficient in deep learning frameworks including PyTorch and TensorFlow.
- Strong understanding of modern deep learning architectures, including Transformer, Self-Attention, Seq2Seq, CNN, BERT, and GPT-based models.
- Experienced with Large Language Models (LLMs), including ChatGPT, ChatGLM, LLaMA, Qwen, and related open-source LLM ecosystems.
- Deep understanding of LLM architecture, pre-training, instruction tuning, alignment, inference optimization, and deployment.
- Experienced in NLP tasks including named entity recognition, relation extraction, machine reading comprehension, knowledge graphs, semantic matching, and text generation.
- Skilled in parameter-efficient fine-tuning methods, including LoRA, Prefix-Tuning, Prompt-Tuning, and instruction tuning.
- Experienced with distributed training and inference acceleration frameworks, including DeepSpeed and vLLM.
- Familiar with multimodal AI models, including Vision Transformer (ViT), CLIP, and vision-language models.
- Experienced in containerized AI development and deployment using Docker.
- Knowledgeable in reinforcement learning algorithms, including DQN, PPO, Actor-Critic, RLHF, and preference optimization methods such as DPO.

📊 STRENGTHS

- **Data-driven Decision Making:** Strong ability to analyze data, design experiments, and make technical decisions based on quantitative evaluation.
- **Cross-domain AI Integration:** Experienced in applying AI technologies to different industries, including automotive, telecommunications, and enterprise applications.
- **Research-to-Production Capability:** Ability to transform academic research achievements into scalable industrial AI solutions.
- **Innovation and Problem Solving:** Skilled at identifying key problems in complex business scenarios and designing effective algorithmic solutions.

PROJECT EXPERIENCE

Project 1: Intelligent Cockpit Voice Interaction System

Nov. 2024 – Present

Technology Stack: Python + RAG + LangChain + FastAPI + VAE + BERT + Qwen3 + vLLM

Project Overview:

Developed an intelligent cockpit voice interaction system for automotive applications, aiming to provide accurate and efficient voice-based interaction capabilities across navigation, vehicle control, multimedia, and other scenarios.

Core Algorithm Contributions:

- **Small Model-based Intent Filtering and Understanding**
 - Designed a VAE-based lightweight filtering module to identify and remove invalid voice commands, significantly reducing unnecessary LLM inference costs.
 - Developed BERT-based models for multi-class classification, multi-label classification, and sequence labeling tasks to accurately identify user intents and extract key entities.

- Built a verification module to compare original user queries with generated execution functions, improving system reliability and preventing incorrect command execution.
- **LLM-based Semantic Understanding and Task Planning**
 - Developed semantic rewriting modules to transform ambiguous user commands into structured representations for downstream processing.
 - Designed complex instruction decomposition strategies to split multi-intent commands into independent executable subtasks.
 - Participated in the development of Planner modules, routing tasks into different domains including navigation, vehicle control, and multimedia.
 - Optimized generation modules with Retrieval-Augmented Generation (RAG) and Few-shot Learning, enabling dynamic execution function generation using historical data and retrieved knowledge.

Deployment Results:

- Lightweight filtering model eliminated more than 99% of invalid requests, significantly reducing computational overhead.
- LLM-based semantic understanding improved complex instruction recognition accuracy to over 98%.
- RAG-enhanced Few-shot Learning improved new scenario adaptation efficiency by 40%.

Project 2: LLM-based SQL Intelligent Assistant

Jun. 2023 – Apr. 2024

Technology Stack: Python + ChatGLM + Rasa + PyTorch + Flask + Docker

Project Overview:

Developed an intelligent SQL assistant based on Large Language Models, enabling users to query databases through natural language. The system converts user requirements into executable SQL statements and provides intelligent database interaction capabilities.

Core Algorithm Contributions:

- Conducted comprehensive benchmark experiments on Spider and BIRD datasets, comparing multiple LLM architectures including ChatGLM, Vicuna, and LLaMA.
- Fine-tuned the optimal LLM using LoRA and instruction tuning strategies. Designed dynamic prompt template selection and curriculum learning methods to improve SQL generation accuracy.
- Designed a hierarchical dialogue architecture based on Rasa, combining NLU models and rule-based domain management. Decomposed business scenarios into 26 sub-domains, achieving 99.98% SQL generation accuracy in practical applications.
- Addressed engineering challenges in LLM deployment, including date/time normalization and complex temporal information extraction. Developed joint fine-tuning strategies for SQL generation and temporal understanding, enabling cross-domain model reuse.

Performance Optimization:

- Applied FastLLM inference acceleration techniques, reducing response latency from 7 seconds to 2 seconds and improving QPS by 3.5×

Project 3: Telecom Alarm Knowledge Graph and Frequent Pattern Mining System

Jan. 2023 – May 2023

Technology Stack:

Python + FP-Growth + Apriori + Neo4j

Project Overview:

Developed an intelligent telecom operation and maintenance analysis system to discover hidden relationships among massive network alarms. The system combined frequent pattern mining algorithms and knowledge graphs to support automated fault analysis and diagnosis.

Core Algorithm Contributions:

- Modeled telecom alarm events and designed sliding-window temporal segmentation algorithms to transform approximately 4 million daily alarm records into structured transaction datasets.

- Improved FP-Growth algorithm with distributed parallel computation strategies for multi-region and multi-datacenter alarm mining, improving frequent pattern mining efficiency by 50%.
- Introduced dynamic minimum support adjustment strategies according to alarm categories, improving expert validation acceptance rate to 92%.

Deployment Results:

- Reduced redundant alarm transmission by 18% in transmission networks and 15% in wireless networks, eliminating approximately 530,000 redundant alarms per day.
- Developed Flask-based APIs for real-time knowledge graph querying and reduced fault localization time by 65%.